GOKUL GOPALAKRISHNAN

AI PRODUCT MANAGER / TECHNICAL PROGRAM MANAGER / PRODUCT ENGINEER

github.com/gokulg846

POSITIONING

Product and program operator at the seam between data/ML systems and physical hardware. Frames architecture, latency, evaluation, bandwidth, and reliability tradeoffs as measurable product decisions across AI governance, inspection, telemetry, and edge systems.

EXPERIENCE

RisingPhoenix.ai
JUL 2026 - PRESENT

AI PRODUCT ENGINEER INTERN

Scoped evaluation rigor, traceability, and release gates for an AI-native governance pipeline spanning evidence ingestion and compliance mappings.

Cummins
FEB 2025 - JAN 2026

PRODUCT ENGINEER CO-OP

Connected physical-system data to engineering decisions through structured analysis and technical dependency ownership.

Purdue-Bayer Innovation
AUG 2023 - DEC 2023

DATA SCIENCE CONSULTANT

Translated an applied data-science problem into an evaluation plan, analytical workflow, and decision artifact.

Accenture
JUL 2021 - JUL 2023

ANALYST, DATA & ANALYTICS

Built and operated IIoT streaming and predictive-maintenance workflows across 10,000+ sensors.

EDUCATION

Purdue University
Master of Engineering Management · May 2026

SASTRA University
B.Tech, Mechanical Engineering

SELECTED SYSTEMS

NCR-Bench / AI evaluation
Delayed agent scaling by three weeks to establish a locked benchmark, severity gates, and traceable evidence. **94.8% reliability; 80% fewer manual audit cycles; zero critical hallucinations in production testing.**

Industrial telemetry platform
Idempotent raw-to-mart pipeline using S3/MinIO, Postgres, dbt, Prefect, Streamlit, Docker, and Terraform, with source-freshness and schema-quality gates.

Semiconductor yield analytics
Causal wafer simulation and medallion workflow covering about 943k die records, 139 dbt tests, spatial wafer maps, SPC signals, Pareto loss views, and DuckDB marts.

Sensor anomaly evaluation
Leakage-aware CWRU vibration pipeline comparing Isolation Forest and an LSTM autoencoder trained on healthy data, with a fixed 0.5% false-positive budget.

SYSTEMS TOOLKIT

- ML evaluation: LLM benchmarks, grounding, precision/recall, threshold calibration, slice analysis
- Data systems: Python, SQL, PyTorch, Kafka/Redpanda, dbt, Prefect, FastAPI, WebSockets
- Edge and hardware: sensor telemetry, compute/latency budgets, HITL gates, failure-aware fallbacks
- Program execution: requirements, dependency maps, API contracts, go/no-go gates, rollback criteria

DECISION RECORDS

Inspection threshold policy
Set a 0.94 critical-path recall floor and accepted +4 percentage points of review load; 12 ms edge inference and zero critical leakage on the locked gate.

Adaptive telemetry
Specified 100 Hz transmission for physical state changes and 1 Hz after stable idle; architecture target exceeds 10,000 messages/second at <50 ms map latency.

AMR edge AI - concept
Proposed local perception and stopping with server-side static maps; targets <=22 ms p95 perception and 99.99% stop-request reliability pending hardware validation.

PUBLIC CODE

github.com/gokulg846/data_telemetry
github.com/gokulg846/Semiconductor-wafer-yield-analysis-pipeline
github.com/gokulg846/industry-sensor-anomaly-detection
github.com/gokulg846/AI-Continuous-Compliance